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3-Hour NVMe SSD Validation Report

2x Samsung PM1743 15.36 TB (Gen5) + 2x 960 GB NVMe

Server / Host	172.16.13.217 (172.16.13.217)
Platform	Tyrone MDA200A2N-224 / 2x AMD EPYC 9135 (32C/64T), 1.13 TB DDR5
Validation Tier	Qualification
Campaign Window	02 Jun 2026 06:30-06:45 UTC
Prepared For	Netweb Technologies India Limited
Date	2026-06-02

Overall Result: PASS WITH LIMITATIONS

Executive Summary

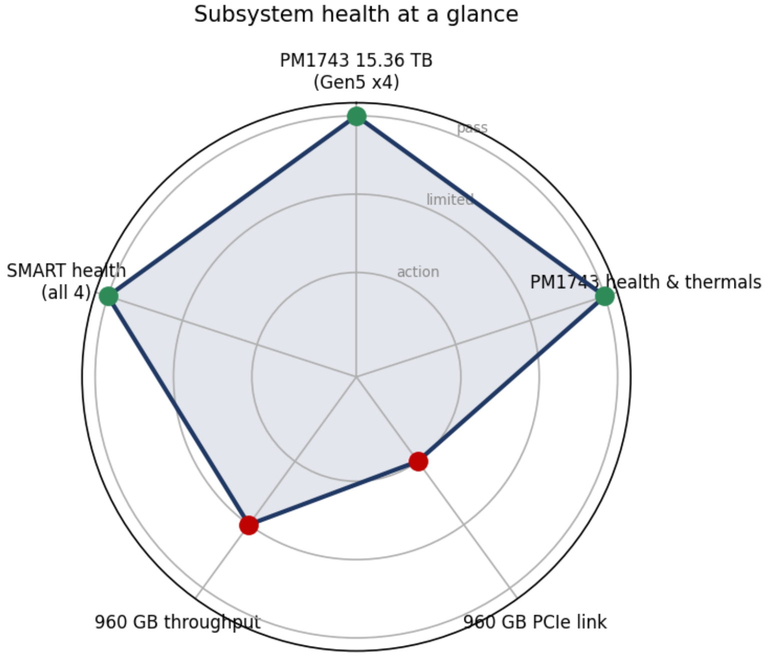
Performance validation of four Samsung NVMe SSDs in server 172.16.13.217 (Tyrone MDA200A2N-224, dual AMD EPYC 9135): two 15.36 TB Samsung PM1743 enterprise U.2 drives (PCIe Gen5) and two 960 GB Samsung M.2 client drives. Each drive was exercised with a NUMA-pinned fio 3.28 direct-IO suite (sequential 1 MiB, 4 KiB random at QD512, mixed 70/30, and QD1 latency), one-per-socket in two waves to eliminate cross-drive contention.

Both PM1743 drives are validated for production at full PCIe Gen5 x4: ~12.3 GB/s sequential read, 7.06 GB/s write, ~1.75M 4K random-read IOPS at QD512, and 12 us QD1 write latency, with perfect health.

The two 960 GB drives are functionally healthy but severely throughput-limited: their PCIe link negotiated at Gen3 x2 instead of Gen4 x4 - a double downgrade - capping them at ~1.87 GB/s read and ~443K IOPS (the 1.8 GB/s x2 ceiling), about 26% of capability. This is a platform/slot link-training fault, not a drive defect, and must be corrected before these drives are qualified.

Validation Scorecard

Subsystem	Result	Detail	Verdict
PM1743 15.36 TB (Gen5 x4)	12.3 GB/s read, ~1.75M IOPS	Full Gen5 x4; 7.06 GB/s write; 12 us QD1 write	PASS
PM1743 health & thermals	0 errors, 0% wear, <=32 C	No thermal throttling under load	PASS
960 GB PCIe link	Gen3 x2 (double downgrade)	Gen4 x4 capable; trained 1 gen + half lanes too low	ACTION
960 GB throughput	1.87 GB/s read, 443K IOPS	Link-capped at the 1.8 GB/s x2 wall (not NAND)	LIMITED
SMART health (all 4)	0 media errors, 0 crit warnings, 0% wear	Clean	PASS



Hardware Inventory

System

Vendor	Tyrone Systems
Model	MDA200A2N-224
Board	MH12XM

Chassis	2U
Hostname	172.16.13.217
Serial	n/a

CPU

Model	2x AMD EPYC 9135
Sockets	2
Cores Per Socket	16
Threads	64
Numa Nodes	2
Numa Device Map	PM1743 (nvme0/1) -> node1; 960 GB (nvme2/3) -> node0

Memory

Total	1.13 TB
Type	DDR5

BIOS / Platform

Version	AMI ES312AMS.205T8
Date	2026-03-26

BMC / Power

Firmware	1.08

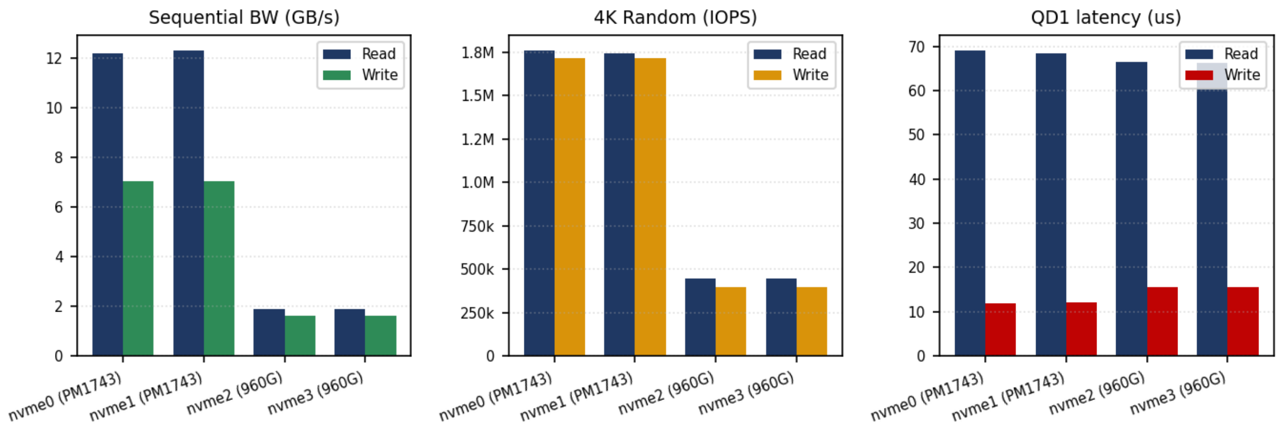
Kernel / OS

Distro	Ubuntu 22.04.4 LTS
Kernel	6.8.0-117-generic

Storage Devices

Dev	Model	Form	Capacity	Serial	Link (neg)	Link (cap)
/dev/nvme0n1	Samsung PM1743	U.2 2.5in	15.36 TB	n/a (FW LDDM7U2Q)	Gen5 x4	Gen5 x4
/dev/nvme1n1	Samsung PM1743	U.2 2.5in	15.36 TB	n/a (FW LDDM7U2Q)	Gen5 x4	Gen5 x4
/dev/nvme2n1	Samsung PM9-series	M.2 22x80	960 GB	n/a (FW LDDD2U2Q)	Gen3 x2	Gen4 x4
/dev/nvme3n1	Samsung PM9-series	M.2 22x80	960 GB	n/a (FW LDDD2U2Q)	Gen3 x2	Gen4 x4

Storage Performance

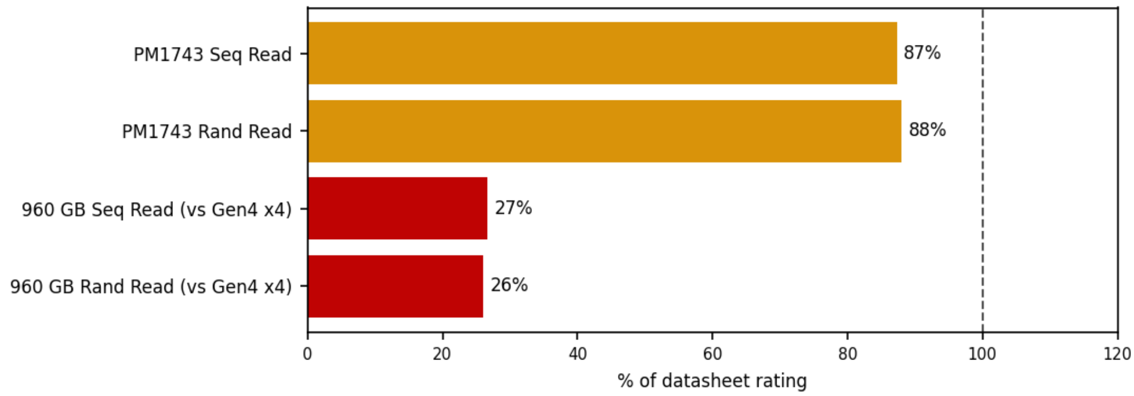


Drive	Link	SeqR	SeqW	RandR	RandW	QD1R	QD1W	p99.9R
nvme0 (PM1743)	Gen5 x4	12.2	7.1	1.76M	1.72M	69	12	n/a
nvme1 (PM1743)	Gen5 x4	12.3	7.1	1.74M	1.72M	68	12	n/a
nvme2 (960G)	Gen3 x2	1.9	1.6	443k	395k	66	16	n/a
nvme3 (960G)	Gen3 x2	1.9	1.6	443k	396k	66	16	n/a

Units: SeqR/SeqW in GB/s; RandR/RandW in IOPS; QD1 & p99.9 in microseconds.

Specification Compliance

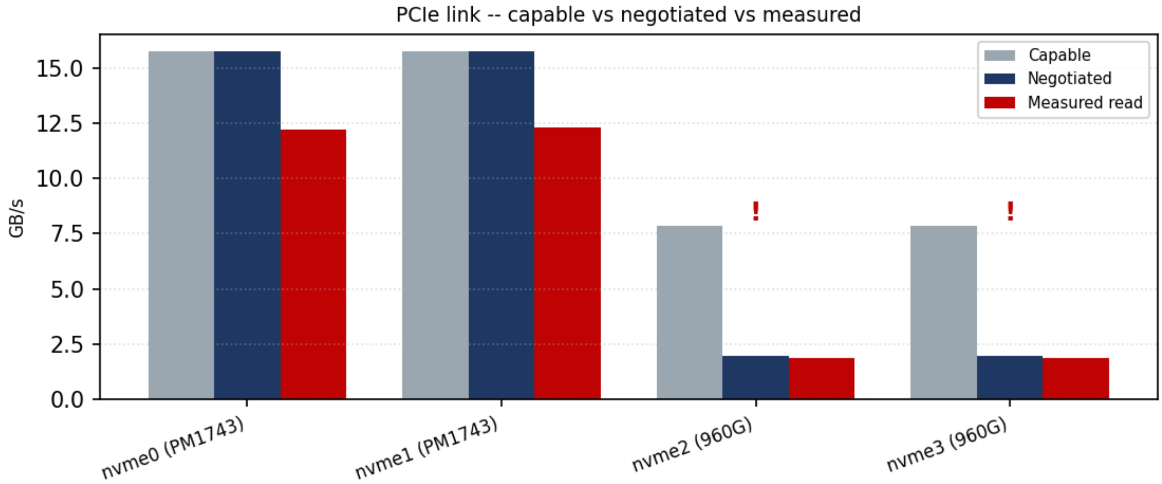
Measured performance as a percentage of datasheet rating. Reference line at 100%. Bars auto-colored: ≥90% green, 75-90% amber, <75% red.



Metric	Measured	Rated	Unit	% of rating
PM1743 Seq Read	12.22	14	GB/s	87%
PM1743 Rand Read	1,760,240	2,000,000	IOPS	88%
960 GB Seq Read (vs Gen4 x4)	1.87	7	GB/s	27%
960 GB Rand Read (vs Gen4 x4)	443,231	1,700,000	IOPS	26%

PCIe Link Analysis

Capable vs negotiated link bandwidth vs measured sequential read. Drives negotiating below their capable width/speed are flagged.



Link warnings: nvme2 (960G): Gen3 x2 negotiated vs Gen4 x4 capable -- double downgrade, ~26% of capability; nvme3 (960G): Gen3 x2 negotiated vs Gen4 x4 capable -- double downgrade, ~26% of capability

Component Health

Drive	Link	Temp pre->post	Wear	Media errors	Verdict
nvme0n1 (PM1743)	Gen5 x4	24->32 C	0%	0	PASS
nvme1n1 (PM1743)	Gen5 x4	24->31 C	0%	0	PASS
nvme2n1 (960 GB)	Gen3 x2	36->42 C	0%	0	PASS
nvme3n1 (960 GB)	Gen3 x2	35->41 C	0%	0	PASS

Kernel log (dmesg) diff: All four drives: 0 media/data-integrity errors, 0 critical warnings, 0% endurance used. No thermal throttling (peak 42 C). 960 GB QD1 tail latency p99 ~3.6 ms vs ~0.47 ms on the PM1743 - a function of the Gen3 x2 link, not the NAND.

Findings & Recommendations

Findings

- PM1743 15.36 TB drives are production-ready: both deliver ~12.3 GB/s read, 7.06 GB/s write, ~1.75M 4K random-read IOPS at QD512, ~1.0M read IOPS in a 70/30 mix, and 12 us QD1 write latency - fully validated at PCIe Gen5 x4.
- All four drives are electrically healthy: zero media errors, zero critical warnings, 0% endurance used, clean thermals (<=42 C) under sustained load.
- The two 960 GB drives are link-throttled, not faulty: both negotiated PCIe Gen3 x2 instead of Gen4 x4, capping throughput at the ~1.8 GB/s x2 wall (1.87 GB/s read, 443K IOPS) - about 26% of capability. QD1 latency stays healthy since single-IO latency is not link-width bound.
- Consistency across pairs is excellent: the two PM1743 drives differ by <1% on every metric, as do the two 960 GB drives - indicating uniform parts and a stable test harness.

Recommendations

- Fix the 960 GB PCIe link (priority): re-seat the drives; verify the M.2/U.2 slot is wired x4; check BIOS PCIe bifurcation for the node0 slots (PCI 51:00 / 52:00); confirm the riser/backplane is not forcing Gen3. Re-run lspci to confirm Gen4 x4 before re-qualifying - expect ~6-7 GB/s and 3-4x the random IOPS.

- Consider 4K LBA format for the PM1743 drives: all namespaces are 512-byte; enterprise NVMe generally performs best at 4096-byte LBA (destructive nvme format before deployment).
- For steady-state endurance numbers, run a full precondition: the reported random-write figures are fresh-state; a 2x full-span fill plus 30+ min random-write soak per drive would yield JEDEC steady-state IOPS.
- Enable continuous monitoring: install nvme-cli / smartd alerting on the deployed host for temperature, media errors and endurance.

Conclusion

Device	Outcome	Verdict
2x Samsung PM1743 15.36 TB (nvme0/1)	Validated at PCIe Gen5 x4, full performance, healthy	PASS
2x Samsung 960 GB (nvme2/3)	Healthy but PCIe link downgraded to Gen3 x2; correct slot/bifurcation, then re-test	LIMITED

Reproducibility Appendix

Exact tooling, kernel command line, BIOS state and a SHA-256 of the raw data bundle, so this run can be audited and reproduced.

Fio	fio 3.28
Nvme Cli	nvme-cli 1.16
Distro	Ubuntu 22.04.4 LTS
Kernel	Linux 6.8.0-117-generic
Bios	AMI ES312AMS.205T8 (2026-03-26)
Bmc Firmware	1.08
Baseboard	MH12XM
Methodology	ioengine=libaio, direct=1 (raw block device), numactl per-drive NUMA node, 2 waves one-per-socket
Campaign Window	02 Jun 2026 06:30-06:45 UTC
Note	Reconstructed from the validated source report into the CoreBench results.json contract.